

Data Cleaning Report

Retail Sales Dataset - Step-by-Step Methodology

1. Basic Cleanup & Setup

Resizing for Visibility

Fixed the row and column sizes so all the data fits perfectly on the screen.

- **Shortcut:** Alt + H + O + A (Autofit Row Height)
- **Shortcut:** Alt + H + O + I (Autofit Column Width)

Removing Empty Rows

Found rows that were completely blank and deleted them.

- **Formula:**

```
=COUNTBLANK( range )
```

- **Action:** Pressed **Ctrl + Shift + L** to open filters, filtered for the blanks, and deleted the entire rows.

Finding Missing Values

Highlighted all empty cells at once to tag them before cleaning.

- **Shortcut:** Ctrl + G > Special > Blanks. Typed NA and pressed Ctrl + Enter to fill them all at once.

2. Fixing Words and Dates

Standardizing Dates

Made sure all the dates were written in the exact same format (DD-MM-YYYY).

- **Formula:**

```
=IF(ISNUMBER(B2), TEXT(B2, "dd-mm-yyyy"), TEXT(DATE(RIGHT(B2, 4),  
MONTH(1&LEFT(B2, 3)), ...), "dd-mm-yyyy"))
```

- **Action:** Copied the new dates and used **Paste as Values (123)** over the old ones.

Product Names & Gender

Cleaned up the names by removing package weights (changed "Rice (1kg)" to just "Rice").

Cleaned up simple spelling formatting.

- **Shortcut:** Ctrl + H (Find and Replace).
- **Action for Products:** Find what: (*) | Replace with: *leave completely blank*.

3. Fixing the Money and Numbers

Shipping Cost Errors

Found shipping costs with minus signs (like -45). Since you can't pay negative money for shipping, converted them to normal positive numbers.

- **Formula:**

```
=ABS(Q2)
```

(The Absolute Value formula strips away the minus sign).

Discounts

Some discounts said "NA". Changed those to 0 because "NA" simply indicates no discount was applied.

- **Shortcut:** Ctrl + H (Find what: NA | Replace with: 0).

Quantity Glitches

Some quantities said 999 or NA. Since those were system glitches, used math to find the real number of items bought.

- **Formula:**

```
=ROUND(Sales_Amount / (Unit_Price * (1 - Discount_Pct)), 0)
```

4. Fixing Shipping Times

Crazy Delays & "NA" Shipping

Changed impossible times (like 100 days) and "NA" missing data to the store average, which was 6 days.

- **Formula:**

```
=MEDIAN(R:R)
```

(Calculated the middle/average shipping time).

- **Action:** Filtered for the bad numbers, typed 6, and pressed **Ctrl + Enter** to change them all at once.

5. Fixing Customer Info

Ages

Cleaned up four different types of demographic errors:

- Removed minus signs from negative ages using

```
=ABS(E2)
```

.

- Replaced fake ages (150, 999) with the average customer age (34) using

```
=MEDIAN(E:E)
```

.

- Deleted ages that were 0 or NA and left them completely blank using **Ctrl + H**.

Customer Satisfaction Scores

Some Customer Satisfaction scores said "NA" because the customer didn't vote. Deleted the "NA" and left the cell completely blank so it wouldn't mess up the final store rating average.

- **Shortcut:** Ctrl + H (Find what: NA | Replace with: *leave completely blank*).

6. Deleting Duplicates

Double Entries

Found rows that the system accidentally copied twice (True Duplicates).

- **Action:** Highlighted data, went to Data Tab > **Remove Duplicates** > Checked "(Select All)" columns to safely delete the extra copies without losing valid multi-item orders.